

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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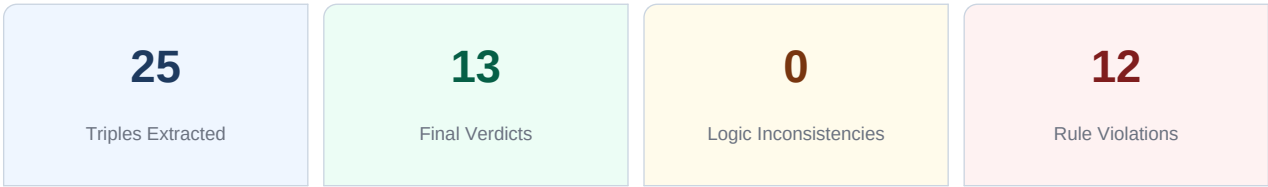
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REPORT OVERVIEW & METHODOLOGY

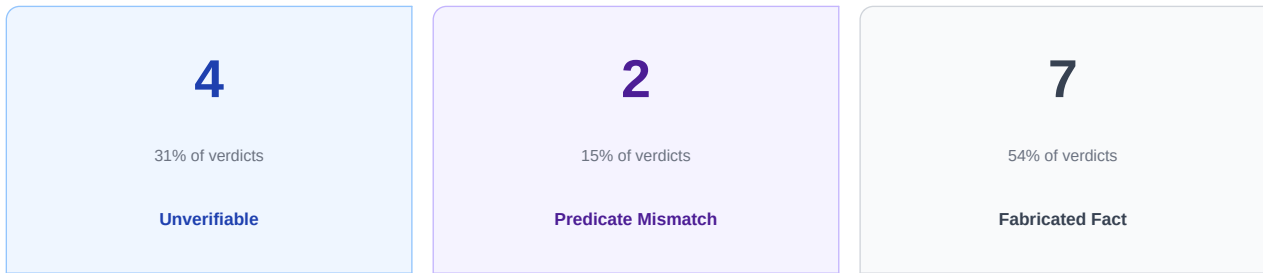
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	HAS_INTEREST_RATE	30		The percentage of interest charged on a loan for this entity.
2	290	HAS_INTEREST_RATE	20		The percentage of interest charged on a loan for this entity.
3	333	HAS_INTEREST_RATE	40		The percentage of interest charged on a loan for this entity.
4	333	HAS_RISK_TIER	high prime		The category of risk assigned to the borrower.
5	290	HAS_RISK_TIER	high prime		The category of risk assigned to the borrower.
6	301	HAS_RISK_TIER	deep subprime		The category of risk assigned to the borrower.
7	333	HAS_APPROVAL_ODDS	200		The likelihood or metric associated with the approval of a loan.
8	290	HAS_APPROVAL_ODDS	30		The likelihood or metric associated with the approval of a loan.
9	890	HAS_APPROVAL_ODDS	40		The likelihood or metric associated with the approval of a loan.
10	290	HAS_COLLATERAL_TYPE	vehicle		The type of asset pledged as security for the loan.
11	500	HAS_COLLATERAL_TYPE	vehicle		The type of asset pledged as security for the loan.
12	600	HAS_COLLATERAL_TYPE	assets		The type of asset pledged as security for the loan.
13	301	HAS_INTEREST_RATE	2		The percentage of interest charged on a loan for this entity.
14	303	HAS_INTEREST_RATE	3		The percentage of interest charged on a loan for this entity.
15	333	HAS_INTEREST_RATE	4		The percentage of interest charged on a loan for this entity.
16	777	HAS_INTEREST_RATE	6		The percentage of interest charged on a loan for this entity.
17	433	HAS_INTEREST_RATE	27		The percentage of interest charged on a loan for this entity.
18	533	HAS_INTEREST_RATE	21		The percentage of interest charged on a loan for this entity.
19	633	HAS_INTEREST_RATE	32		The percentage of interest charged on a loan for this entity.
20	N/A	HAS_NUMBER_OF_FAMILY_MEMBERS_NEEDED_TO_TAKE_A_LOAN	Not specified in the provided context		The number of family members required to secure a loan.
21	300	HAS_DESCRIPTION	A score of 300 represents the most financially distressed and highest-risk borrowers, people who have likely experienced multiple defaults, bankruptcies, repossessions, or long stretches of missed payments.		A textual explanation or definition of the entity.
22	N/A	HAS_MINIMUM_AMOUNT_OF_LOAN_IN_USD	Not specified in the provided context		The lowest possible amount of money that can be borrowed in US Dollars.

#	Subject	Predicate	Object	Type	Predicate Definition
23	N/A	HAS_MAXIMUM_AMOUNT_OF_LOAN_IN_USD	Not specified in the provided context		The highest possible amount of money that can be borrowed in US Dollars.
24	305	HAS_APPROVAL_ODDS	1		The likelihood or metric associated with the approval of a loan.
25	310	HAS_DEFAULT_PROBABILITY	2		The likelihood that the borrower will fail to repay the loan.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
290	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
290	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
301	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
333	HAS_APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
890	HAS_APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_COLLATERAL_TYPE	vehicle	HAS_COLLATERAL_TYPE	HIGH
500	HAS_COLLATERAL_TYPE	vehicle	HAS_COLLATERAL_TYPE	HIGH
600	HAS_COLLATERAL_TYPE	assets	HAS_COLLATERAL_TYPE	HIGH
301	HAS_INTEREST_RATE	2	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
303	HAS_INTEREST_RATE	3	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	4	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
777	HAS_INTEREST_RATE	6	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
433	HAS_INTEREST_RATE	27	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
533	HAS_INTEREST_RATE	21	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
633	HAS_INTEREST_RATE	32	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM

Subject	Original Predicate	Object	Canonical Predicate	Conf.
N/A	HAS_NUMBER_OF_FAMILY_MEMBERS_NEEDED_TO_TAKE_A_LOAN	Not specified in the provided context	HAS_NUMBER_OF_FAMILY_MEMBERS_NEEDED_TO_TAKE_A_LOAN	LOW
300	HAS_DESCRIPTION	A score of 300 represents the most financially distressed and highest-risk borrowers, people who have likely experienced multiple defaults, bankruptcies, repossessions, or long stretches of missed payments.	HAS_DESCRIPTION	LOW
N/A	HAS_MINIMUM_AMOUNT_OF_LOAN_IN_USD	Not specified in the provided context	HAS_MINIMUM_AMOUNT_OF_LOAN_IN_USD	LOW
N/A	HAS_MAXIMUM_AMOUNT_OF_LOAN_IN_USD	Not specified in the provided context	HAS_MAXIMUM_AMOUNT_OF_LOAN_IN_USD	LOW
305	HAS_APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
310	HAS_DEFAULT_PROBABILITY	2	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	HIGH

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	c_interest_rate_range	Violation because subject credit score 890 > 850 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	c_interest_rate_range	Violation because subject credit score 290 < 300 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40	c_interest_rate_range	Violation because object interest rate 40 > 30 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	Violation because subject 333 is between 300 and 579 but object does not contain 'deep subprime'.
factually_inconsistent	RULE_VIOLATION	290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_range	Violation because subject credit score 290 < 300 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	c_approval_odds_range	Violation because object approval odds 200 > 100 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	c_approval_odds_range	Violation because subject credit score 290 < 300 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	c_approval_odds_range	Violation because subject credit score 890 > 850 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	290	HAS_COLLATERAL_TYPE	vehicle	c_collateral_type	Violation because subject credit score 290 < 300 which is out of allowed range.
factually_inconsistent	RULE_VIOLATION	633	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	32	c_interest_rate_range	Object 32 exceeds maximum interest rate of 30, violating constraint c_interest_rate_range.
factually_inconsistent	RULE_VIOLATION	305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	c1_approval_odds	Subject 305 is in 300-345 range requiring object to semantically contain '8', but object is '1', violating c1_approval_odds.
factually_inconsistent	RULE_VIOLATION	310	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	2	c1_default_prob	Credit score 310 is between 300 to 395 with incremental of 5 points but object 2 does not semantically contain 38.5

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Fabricated Fact	factually inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●●●●●●● ●●● 1.00	Predicate and object match entries in DB but the subject '301' does not appear in DB candidates for this predicate-object pair. Ev: subject: 300, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 305, predicate: RISK_TIER_CLASSIFIED_AS_B...
500	HAS_COLLATERAL_TYPE	vehicle	Predicate Mismatch	factually inconsistent	PREDICATE_MISMATCH	STRICT	PREDICATE	●●●●●●■ ■■■ 0.60	Subject '500' exists in DB but 'HAS_COLLATERAL_TYPE' is not a known predicate for it. Ev: subject: 500, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 500, predicate: INTEREST_RATE_CHARGED_AS...
600	HAS_COLLATERAL_TYPE	assets	Predicate Mismatch	factually inconsistent	PREDICATE_MISMATCH	STRICT	PREDICATE	●●●●●●■ ■■■ 0.60	Subject '600' exists in DB but 'HAS_COLLATERAL_TYPE' is not found as predicate. Ev: subject: 600, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: subprime; subject: 600, predicate: INTEREST_RATE_CHARGED_AS_ANNUA...
301	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2	Fabricated Fact	factually inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●■■■■■ ■■■ 0.20	Predicate is known, but no entries with subject '301' and this predicate exist in DB.
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	3	Fabricated Fact	factually inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●■■■■■ ■■■ 0.20	Predicate is known, but no entries with subject '303' and this predicate exist in DB.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●■■■■■ ■■■ 0.20	Predicate is known, but no entries with subject '333' and this predicate exist in DB.
777	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	6	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●■■■■■ ■■■ 0.20	Predicate is known, but no entries with subject '777' and this predicate exist in DB.
433	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	27	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●■■■■■ ■■■ 0.20	Predicate is known, but no entries with subject '433' and this predicate exist in DB.
533	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	21	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	STRICT	SUBJECT	●●■■■■■ ■■■ 0.20	Predicate is known, but no entries with subject '533' and this predicate exist in DB.
N/A	HAS_NUMBER_OF_FAMILY_MEMBERS_NEED_TO_TAKE_A_LOAN	Not specified in the provided context	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is unknown in DB schema and no matching DB entries exist.
300	HAS_DESCRIPTION	A score of 300 represents the most financially distressed and highest-risk borrowers, people who have likely experienced multiple defaults, bankruptcies, repos sessions, or long stretches of missed payments.	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is unknown and there is no DB evidence for subject or object.
N/A	HAS_MINIMUM_AMOUNT_OF_LOAN_IN_USD	Not specified in the provided context	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is unknown and there is no DB evidence for subject or object.
N/A	HAS_MAXIMUM_AMOUNT_OF_LOAN_IN_USD	Not specified in the provided context	Unverifiable	unverifiable	—	NONE	—	■■■■■■■ ■■■ 0.00	Predicate is unknown and there is no DB evidence for subject or object.

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.